Predicting Song Popularity Through Machine Learning and Sentiment Analysis on Social Networks

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Abstract. In this paper, we delve into the intricate relationship between technology, music, and success. Our research focuses on leveraging the capabilities of machine learning algorithms to forecast the success and popularity of songs. By combining audio features, social media data, and emotion analysis, our machine learning-based model demonstrates its capability as a reliable tool for artists, amateurs, and music industry stakeholders to make informed and targeted decisions. Our analysis encompassed not only the inherent audio characteristics of each track but also the extent of audience engagement and emotional response of the fans. Through the integration of textual emotion analysis, we quantified the reactions of the audience and performed an emotion labelling, leveraging our social information to uncover the emotions surrounding each track's release. The experimental results show that the feature data from which the classification algorithms were trained, unfolded to be qualitative and precise, while they also show that the use of the social media features improves the classifiers' performance in predicting the popularity of the tracks.

Keywords: Audio Features · Data Analysis · Emotion Recognition · Social Networks · Supervised Machine Learning

1 Introduction

Music has perennially served as one of the most captivating means for individuals to utilize their leisure time, emphasizing its enormous significance in shaping human experiences and cultural landscapes. Recent data from the IFPI's Global Music Report reveals that worldwide recorded music revenues reached a remarkable sum of 26.2 billion in 2022, reflecting a notable 9% increase compared to the preceding year's total of 24 billion, signifying the eighth consecutive year of growth for the global music industry following a prolonged period of gradual

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decline [1]. Consequently, artists have experienced an acceleration in the production of musical compositions, owing to advancements in technology and the facilitation of artificial intelligence. With the vast availability of digital platforms and social networks, artists and industry stakeholders face a new set of challenges and opportunities in predicting song popularity, developing a need to strike a balance between artistic creativity and commercial success.

This research is about to identify a recurring sequence amidst music, artificial intelligence, and ubiquity in the modern era. It explores the potential of machine learning algorithms in predicting the popularity of songs, presenting a methodology that bridges the gap between artistic intuition and data-driven decision-making. Drawing upon a diverse dataset comprising audio features and social data, we seek to develop a reliable tool for predicting the success and popularity of songs. The use of musical features allows for a comprehensive analysis of the inherent characteristics of each track, while social data mined from platforms, such as X (formerly known as Twitter), provides valuable insights into audience engagement and emotional response. By quantifying the emotional sentiment surrounding a track, we can gain a deeper understanding of its reception and likeability for success. Through the integration of machine learning algorithms, our research offers the potential for accurate predictions of a track's position in popular charts or rankings. This predictive model has the potential to transform decision-making processes in the music industry, enabling music professionals to make well-informed choices that integrate artistic intuition with quantitative analysis.

The subsequent sections of this paper are structured as follows. Section 2 provides an overview of existing research on applications of song classification. In Sect. 3, we present a machine learning model that leverages a combination of audio features, social media indicators, and emotional analytics to accurately predict song popularity. Section 4 presents the experimental results, including an analysis of the most significant features in our methodology and a comprehensive evaluation of the classification performance across different experiments, by emphasizing the value of social media features in predicting song popularity. Finally, we conclude the paper by summarizing our findings and outlining potential avenues for future research in Sect. 5.

2 Related Work

In the related work, several studies have investigated the relationship between audio features and song popularity. One such study [8] emphasizes this connection, while another one [17] explores the correlations between track popularity and key characteristics derived from the Spotify API, with danceability being identified as a significant factor. However, it should be emphasized that no definitive formula has been established that can accurately determine the precise relationship between audio features and the level of popularity. In a different approach, valuable insights are provided into the dynamic nature of song likability over time, shedding light on the evolution of the concept and the emergence of new popular songs in different eras [10]. Furthermore, a recent paper [6]

highlights the importance of audio features such as danceability, liveness, and speechiness in determining song success, and suggests the potential benefits of incorporating lyrics and their specific characteristics such as keywords, length, and language to enhance predictive models and gain deeper audience insights.

Many interesting studies exist regarding how much the lyrics and their content can affect the success of a song, through the process of creating emotions for the listeners. The field of music emotion recognition has extensively explored the detection of emotions conveyed through audio signals [23], employing deep learning algorithms to recognize music emotions through dimensional models [25]. Moreover, the influence of lyrics on audience sentiment has also been investigated, demonstrating its connection to the emotional response to music [21], while the combination of audio features, lyrics, and temporal information has been shown to be effective in predicting song success [24]. In the context of social networks, emotions analysis has been leveraged to explore users' sentiments [18,19], along with various other factors related to user influence [11]. However, it should be emphasized that, to the best of our knowledge, in the context of song predictions the integration of the users' emotions in the social media and the audio features is an area that has been limited explored.

Another perspective that has to be taken into consideration, is also the approval of fans in determining the success of songs. In [2], the authors explored the perceptible "noise" generated among audiences and investigate the impact of variables such as the number of tweets, retweets, comments, and quotes on the top 100 tracks of 2019, as determined by the Billboard chart. Furthermore, the artist-superstar parameter adds an intriguing dimension, as highlighted in [10]. Their findings demonstrate that incorporating non-acoustic features, such as the "superstar" status of an artist, significantly improves the predictive accuracy of song success by 12% compared to models based solely on acoustic features. This finding emphasizes the pivotal role of an artist's fame and popularity in conjunction with the musical characteristics of a song. Unlike traditional techniques that solely focus on either audio or social media indicators and emotions, our research extends the current literature by integrating existing approaches exploring all the main factors that influence the success of a song. By considering all those diverse characteristics, the methodology provides valuable insights in recognizing the song's characteristics that may influence its charts position. Furthermore, it enables a more in-depth and thorough understanding of the relative importance of each factor in determining audience engagement and, consequently the popularity of each song.

3 Model Overview

In Fig. 1, we present our proposed methodology. The process begins with the data collection phase from social networks and music platforms, such as X and Spotify in our case. Once the data have been collected (Sect. 4.2), several data pre-processing techniques are applied on them in order to filter out the noisy data and ensure the data quality (Sect. 4.3). Thereafter, we proceed with feature selection by mining audio characteristics, along with a set of social media

indicators and emotional analytics (Sect. 4.4). Additionally, we conduct several experiments exploring the feature importance, we evaluate the performance of various machine learning classifiers, and finally we proceed with the quality evaluation of the predicted classes (Sect. 4.4). Further technical details about each phase is provided in the following sections.

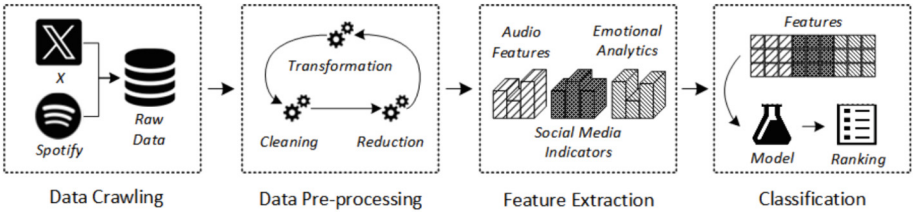

Fig. 1. Overview of the proposed methodology on predicting songs popularity

3.1 Exploring Factors Influencing Song Popularity

In the context of our classification modeling, we employ a comprehensive set of diverse user features to effectively capture the popularity of songs. These features encompass various aspects, including audio characteristics, social media indicators, and linguistic analytics. The audio features provide insights into the musical composition and qualities of the songs, while the social media features help capture public preferences and trends related to the songs. To enrich the classification modelling, we also use the NRC emotion lexicon [13,14], augmented with the WordNet synonyms, to extract the eight basic emotions of the Plutchik's wheel (anger, anticipation, disgust, fear, joy, sadness, surprise and trust) [16]. Thus, each song s is modeled as a tuple, denoted as F_s, which consists of a set of diverse characteristics. These characteristics encompass the overall audience popularity, denoted as $F_s(f_1, f_2, \ldots, f_k)$, where in our case $k = 25$, and are composed of the extracted audio features, social media indicators, and emotional analytics. By considering these diverse aspects, our classification modeling endeavors to provide a comprehensive understanding of the songs and their popularity among the audience.

Capturing Songs' Essence Through Audio Features. The audio features extracted from the songs play a crucial role in the classification process. These features capture various musical aspects and provide insights into the songs' characteristics. By leveraging these features, we can gain a deeper understanding of the musical composition of the songs. Each feature contributes valuable information for classifying songs based on their musical attributes. Below we present the key audio features that were used in our methodology.

- **Danceability**: A measure indicating the suitability of a track for dancing, based on a combination of musical elements (rhythm, beat, and tempo).
- **Energy**: A perceptual measure representing the intensity and activity level of a track.
- **Key**: The tonal center or fundamental pitch upon which a musical composition is constructed, indicating the chosen root note and scale utilized.
- **Loudness**: The overall loudness of a track in decibels (dB), reflecting its perceived acoustic intensity within the audio signal.
- **Speechiness**: An indicator of the presence of spoken words in a track, with higher values suggesting a higher proportion of spoken content.
- **Acousticness**: A measure of the amount of acoustic or non-electronic sound elements in a track, with higher values indicating a more acoustic or "unplugged" sound.
- **Instrumentalness**: The extent to which a track primarily comprises instrumental sounds, devoid of vocal content.
- **Tempo**: The estimated tempo of a track in beats per minute (BPM), representing its speed or pace.
- **Liveness**: A detection of audience presence in a live recording.
- **Valence**: An evaluative assessment of the track's emotional tone, spanning from negative to positive.
- **Artist Followers**: The number of followers an artist has on a particular platform, which reflects the popularity and reach of the artist.

Leveraging Social Media for Predicting Song Popularity. Our approach to classifying songs brings innovation by using social network data to predict their popularity. We look at metrics from social media platforms, such as the communication, amplification, engagement, and commentary, along with the emotional perspective of users' posts to understand the people's views and preferences on specific songs. These indicators provide insights into the level of users' engagement and interaction with songs. In more specific, the following social media key indicators were considered.

- **Communication**: Measures the number of posts of a song, indicating its overall presence and discussion on a social media platform. In the case of the X, this corresponds to the number of tweets mentioning the song.
- **Amplification**: Refers to the extent of sharing and spreading of posts related to a song. On X, this is captured by the number of retweets.
- **Engagement**: Evaluates the level of user interaction and involvement with posts mentioning a song, which is captured via the number of likes on X.
- **Commentary**: Captures the extent of users adding their own thoughts or opinions to posts mentioning a song, and can be measured on X by the number of quotes.

In our classification modeling, a combination of lexicon-based analysis and linguistic techniques were employed as well to capture the emotional aspects of songs based on users' posts in the social network. These metrics provide insights

into the emotions expressed by users when discussing or mentioning songs. The analysis includes the eight basic emotions of the Plutchik's wheel, as well as the overall positive and negative emotions. By examining users' posts, we gain a comprehensive understanding of the emotional responses and sentiments of the audience towards songs in the social network. This linguistic analysis allows us to delve into the emotional impact and reception of songs among users, as expressed through their discussions and interactions within the social network.

Our classification process involves using machine learning techniques and evaluating the model's performance to accurately categorize songs into different levels of popularity. By incorporating social network data into our methodology, we enhance our ability to predict song popularity by gaining valuable insights into audience engagement and industry trends. This innovative integration of social network data complements traditional methods and provides a more comprehensive understanding of song popularity.

3.2 Predicting Songs Popularity

The purpose of this work is to categorize songs based on their rankings. Hence, we chose the Billboard chart as a widely known benchmark for evaluating the success and audience reception of songs across various genres. Therefore, let S represent the set of songs to be classified, and C represent the set of classes. In this case, $C = C_1, C_2, ..., C_k$ represents k distinct classes. Thus, we define a function $f : S \rightarrow C$ that assigns each song s in S to one of the k ranking classes in C, where the function f determines the classification of the songs based on their popularity in the Billboard chart ranking. In the next section, the evaluation metrics and techniques employed to assess the performance of our classification model in predicting songs popularity are discussed.

4 Experimental Results

4.1 Implementation

Our methodology was implemented using Python 3.8.0 as the programming language. The audio features were collected from the Spotify API v2.17.1[1], whereas for the acquisition of the necessary X (Twitter) data, a Python wrapper was developed to interact with the official Twitter API[2] v2. The data pre-processing was employed by the Natural Language Toolkit (NLTK) [3] library in Python, while the implementation of supervised learning algorithms involved the utilization of the scikit-learn machine learning library [15].

[1] https://pypi.org/project/spotipy.
[2] https://developer.twitter.com/en/docs/twitter-api.

4.2 Data Description

In order to study the songs popularity, we focused on the Billboard Top-100 chart of 2020. For each song in the chart, we retrieved its audio features from Spotify. Additionally, we conducted a thorough crawl of posts derived from X, targeting those where the song name was identified as a keyword and the respective artist was mentioned as well. This approach allowed us to filter out noisy tweets that solely included the song title, ensuring a more relevant dataset. The collection process encompassed English-language tweets spanning approximately one year, from November 23, 2019, to November 14, 2020, aligning with Billboard's year-end music recaps. The resulting dataset, after the pre-processing phase, consisted of 9.129.065 retrieved tweets, with a standard deviation of 182.073 tweets per song.

4.3 Data Pre-processing

After the completion of data collection as described in Sect. 4.2, we applied several data pre-processing steps. Our approach involved the elimination of noisy data by excluding user mentions (@*), retweet headers (RT), URLs, punctuation, numbers, and excessive white spaces. We also transformed all textual information to lowercase, tokenized the posts, and removed stop words. Additionally, we removed the outliers from our data to ensure the data integrity. Thus, in our case two songs, #38 and #61 of the Top-100 Billboard chart, were excluded from the analysis as they exhibited extremely high and low levels of activity in the social network. Furthermore, we normalised the extracted feature values in the range of $[0, 1]$.

4.4 Results

Following the feature extraction phase as described in Sect. 3.1, we explored the feature importance, by applying a feature selection algorithm, and by conducting several experiments using different combinations of feature sets and various machine learning classifiers, in order to assess their performance among them, as well. Specifically, in each experiment, we used the following machine learning algorithms, the AdaBoost [9], Decision Tree [5] and Random Forest [4], due to their well-established effectiveness and extensive application [12,20], even surpassing deep learning models' performance [22], in similar research contexts.

Feature Selection. To enhance the learning accuracy and overall quality of our model, we employed the Recursive Feature Elimination (RFE) technique [7] as a feature selection method, with a Random Forest Regressor as the estimator. By employing this technique, the data dimensionality is reduced, and a subset of the most important features is obtained. In each of our experiments, we retained the Top-20 most significant features.

Figure 2 presents the feature weights resulting from our feature selection process, offering insights into the factors that contribute significantly to predicting

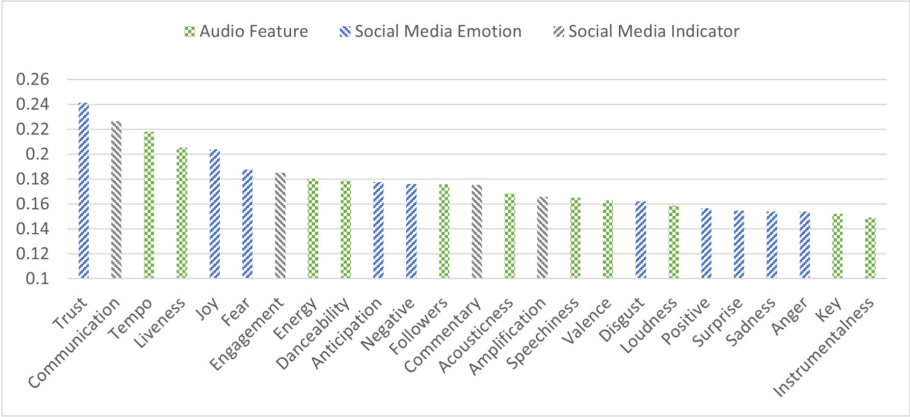

Fig. 2. Feature importance weights for predicting song popularity

the popularity of songs. Among the most significant features, the following key observations can be made. Firstly, social media indicators play a crucial role, as demonstrated by the high weights assigned to features such as communication and engagement. This shows that the level of interaction, discussions, and sharing surrounding a song on social media platforms strongly influences its popularity. Additionally, the presence of specific emotions, such as trust and joy, in social media posts about songs also contributes significantly. Conversely, anger and sadness are considered the least important factors, likely due to their association with songs that may appeal to a more niche audience, encountering greater challenges in achieving mainstream success. These emotional factors provide valuable insights into how people perceive and respond to the music. On the audio side, features like tempo and various acoustic characteristics (e.g., liveness, energy, and danceability) exhibit notable importance. The tempo of a song, for instance, can impact its overall appeal and potential for popularity. Overall, by considering these significant features, our classification model can effectively predict the popularity of songs, leveraging diverse features, by considering both social media aspects and audio characteristics.

Songs Classification. With regard to the song classification, we formulated the problem as a binary classification task, which aims to predict whether a song had the potential to be ranked within the first or second half on the Billboard Top-100 chart. To ensure the reliability of our results, we employed the k-fold cross-validation technique, which involves dividing the dataset into k consecutive subsets and uses the $k-1$ subsets for the model training and the remaining one for the model validation. Specifically, in our experiments, we conducted a 10-fold cross-validation. Moreover, we used the Accuracy and F1-Measure metrics in order to evaluate the performance of the classifier.

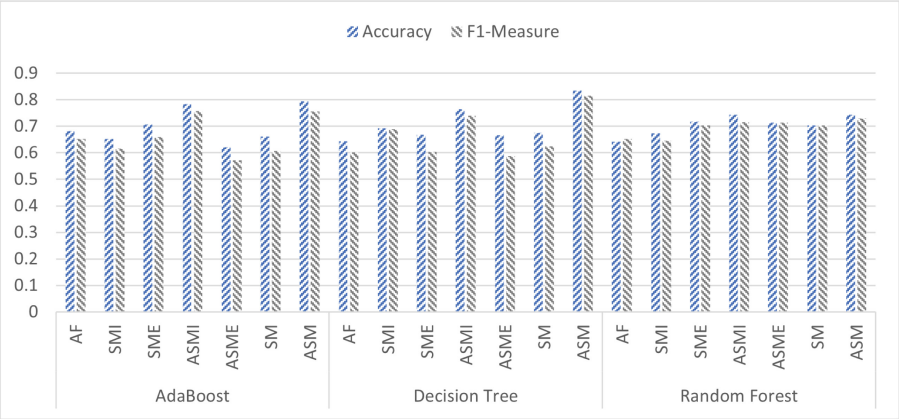

Fig. 3. Evaluating the classifiers accuracy among the different combinations of feature sets

The classification results are depicted in Fig. 3 and provide insights into the performance of different feature combinations and classification algorithms. The experiments were conducted on various feature combinations, by exploring solely each feature category, Audio features (AF), Social Media Indicators (SMI), and Social Media Emotions (SME). Additionally, four hybrid feature combinations were considered, Audio features and Social Media Indicators (ASMI), Audio features and Social Media Emotions (ASME), Social Media Indicators and Social Media Emotions (SM), as well as the combined Audio features and Social Media (ASM) approach. For each feature combination, three well-known classification algorithms were applied, AdaBoost, Decision Tree, and Random Forest.

The classification outcomes demonstrate that the Decision Tree method exhibited superior performance in predicting song popularity, achieving an accuracy score of 0.835 and F1-measure of 0.815 were obtained while employing the ASM combination. This highlights the effectiveness of this algorithm in forecasting song popularity when incorporating both audio features and social media aspects, surpassing other feature combinations. On the other hand, on all the experiments that included Social Media Emotions (SME), the Random Forest model exhibited the optimal performance. Additionally, in scenarios involving Audio Features (AF) and their aggregation with Social Media Indicators (ASMI), the AdaBoost model demonstrates the highest predictive capability. This particular case outscored all the experiments involving either one or two feature categories. Overall, the inclusion of Social Media aspects in conjunction with Audio Features (ASM) proves to be a powerful predictor, with the proposed methodology yielding an approximate 26% increase in classifier accuracy compared to the use of Audio Features alone. This highlights the valuable contribution of integrating Social Media aspects in improving the overall performance of the classifier model.

5 Conclusion

In conclusion, the results obtained from this research are highly remarkable. The algorithm achieved a maximum accuracy of $83,5\%$, demonstrating its ability to predict whether a specific music piece can reach at the top-tier of the annual charts, based on its musical composition and social media interaction on X. As a future work, the utilization of historical Billboard lists to augment the training set size should be considered; however, challenges such as the diminishing relevance of past popularity and large data volumes need to be addressed. The concept of lyrical analysis and its impact on individual emotions and ranking of a song, presents an avenue for investigation. Exploring how various lyrical elements, such as themes, language, rhyming, and sentiment, resonate with listeners and contribute to the overall appeal of a track would provide valuable insights into the dynamics between lyrics, emotions, and hit-making potential. In summary, this paper provides valuable knowledge for the current landscape and lays the foundation for further advancements in predicting music success.

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